Socioeconomic Factors, Charter Status, and High School ACT Scores

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Abstract

This project tests whether socioeconomic factors predict a high school's average ACT score and whether charter schools score differently once those factors are taken into account. I combined 2016–2017 data from EdGap and the National Center for Education Statistics (NCES), which gave 7,227 high schools in 20 states. Using linear regression, I compared single-predictor, quadratic, full, reduced, and charter models with R-squared, mean absolute error (MAE), and ANOVA F-tests. A reduced model with three predictors (the percent of students on free or reduced-price lunch, the percent of adults with a college degree, and the unemployment rate) explained 62.8% of the variation in ACT scores, and its predictions were off by about 1.15 points on average. The percent of students on free or reduced-price lunch was by far the strongest predictor. Charter schools had a lower raw ACT average, but after controlling for the socioeconomic factors the difference was small and not significant.

Introduction

A school's average ACT score is often used as a measure of how well the school is doing. However, schools do not all start from the same place. Some serve students from wealthy neighborhoods, while others serve students from families living in poverty. In this project, I test whether a school's average ACT score can be predicted from socioeconomic factors. I use five socioeconomic measures: the unemployment rate, the percent of adults with a college degree, the percent of children in married-couple families, and the median household income of the census tract around each school, plus the percent of the school's students who qualify for free or reduced-price lunch. As an additional predictor, I also check whether charter schools score differently from non-charter schools after taking these socioeconomic factors into account.

Theoretical Background

The link between family background and school achievement has been studied for decades. It has been found that differences in students' family backgrounds explained more of the differences in achievement between schools than differences in school resources did. The achievement gap between students from high- and low-income families has also grown over time. Based on this research, I expect schools with more low-income students to have lower average ACT scores.

Methodology

The data comes from two sources for the 2016–2017 school year. The first is EdGap (edgap.org), which has 7,986 schools and 7 columns: the NCES school ID, the four census-tract measures, the school's average ACT score (or the ACT equivalent of its SAT score), and the percent of students on free or reduced-price lunch. From the NCES file I kept only the school year, ID, state, ZIP code, school type, school level, and charter status. The notebook (Education_final.ipynb) and the clean data are in my GitHub repository: https://github.com/rockgignac/education.

To prepare the data, I first renamed all columns to short names (for example, rate_unemployment, percent_college, percent_married, median_income, average_act, percent_lunch, and charter). I then did a left merge of EdGap with the NCES data on id, which kept all 7,986 EdGap schools. Next, I checked the range of each numeric column. Two problems showed up: the minimum ACT average was −3.07, and the minimum free lunch percentage was −0.05. Neither is possible, so I set any ACT average below 1 and any free lunch percentage below 0 to missing. Since the ACT is taken by high school students, I kept only schools where school_level equals "High," which also removed the 88 unmatched schools and left 7,230 high schools. I dropped the schools with a missing ACT average rather than filling them in, since a made-up outcome would bias the results. The five numeric predictors still had between 11 and 20 missing values each, so I filled these in with scikit-learn's IterativeImputer using its default settings, fit on the five numeric predictors only. The final clean data set has 7,227 high schools and 13 columns with no missing values. A map of schools by state showed that the data only covers 20 states, with Texas (913), Ohio (654), Illinois (564), and Pennsylvania (543) having the most schools.

For the analysis, I used linear regression with average ACT score as the outcome, fit with the statsmodels formula. First, I fit simple linear regressions with one predictor at a time and checked the fit with residual plots. Because the median income residuals showed a curve, I also fit a quadratic model by adding the term I and compared it to the linear model with an ANOVA F-test. Second, I fit a multiple linear regression with all five socioeconomic predictors. Third, I fit a reduced model with only the predictors that were significant at the 0.05 level and compared it to the full model using R-squared, mean absolute error (MAE), and an ANOVA F-test. Fourth, to compare the size of each predictor's effect, I standardized the reduced model's predictors to a mean of 0 and standard deviation of 1 with StandardScaler and refit the model. Finally, I added charter status to the reduced model as a categorical variable (C(charter)), with non-charter schools as the baseline, and compared it to the reduced model with an ANOVA F-test. MAE was calculated on the full data set.

Computational Results

Single predictors

The percent of students on free or reduced-price lunch had the strongest relationship with ACT scores. Schools with almost no students on free lunch average about 24 on the ACT, while schools where nearly every student qualifies average about 15.5. Median income alone explained only 21.1% of the variation in ACT scores (MAE of 1.71 points), with each additional $10,000 in income linked to about 0.47 more ACT points. The residual plot showed the linear model predicted scores that were too high for the highest-income schools. The quadratic model was a significant improvement, and the negative squared term shows the effect of income levels off at high incomes. However, the R-squared only rose from 0.211 to 0.219 and the MAE only dropped by 0.016 points, so the improvement was very small in practice (Figure 1).

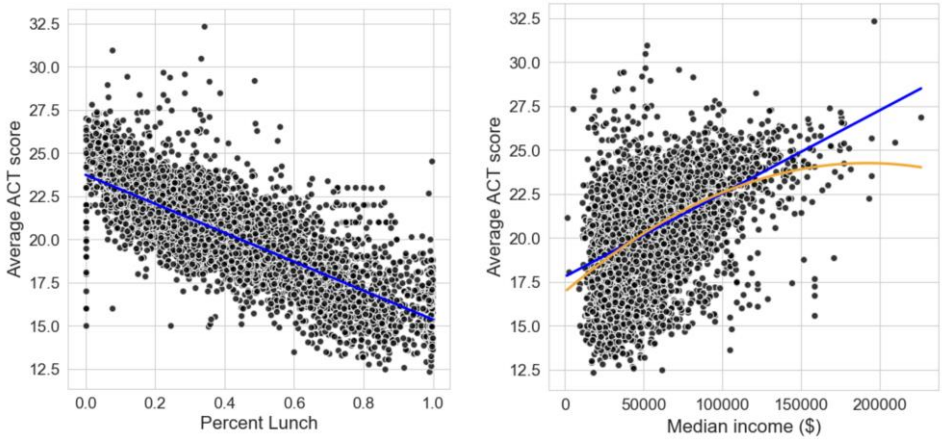


Figure 1. Average ACT score versus the percent of students on free or reduced-price lunch (left) and median household income (right). The blue lines are linear fits and the orange curve is the quadratic fit.

Multiple regression

The model with all five socioeconomic predictors explained 62.8% of the variation in ACT scores, with an MAE of 1.145 points. Free or reduced-price lunch, percent college, and unemployment rate were all significant at the 0.01 level. Percent married and median income were not significant once the other predictors were included. Income was significant by itself, but it overlaps so much with percent college and free lunch that it adds no new information.

Reduced model

Removing percent married and median income did not make the model worse. The ANOVA F-test comparing the two models gave a p-value of 0.81, and the fit was almost identical (Table 1):

Table 1. Comparison of the regression models by number of predictors, R-squared, and mean absolute error (MAE).

Model	Predictors	R-squared	MAE
Median income only	1	0.211	1.713
Median income (quadratic)	2	0.219	1.697
Full model	5	0.628	1.145
Reduced model	3	0.628	1.146
Reduced + charter	3 + charter	0.630	1.140

The reduced model's coefficients were −7.59 for free lunch, 1.71 for percent college, and −2.17 for unemployment, all with $p < 0.001$. Because these variables are proportions from 0 to 1, the free lunch coefficient means that going from 0% to 100% of students on free or reduced-price lunch is linked to an ACT average about 7.6 points lower. The condition number dropped, so the multicollinearity problem was gone. The residuals were spread evenly around 0 with no clear pattern (Figure 2).

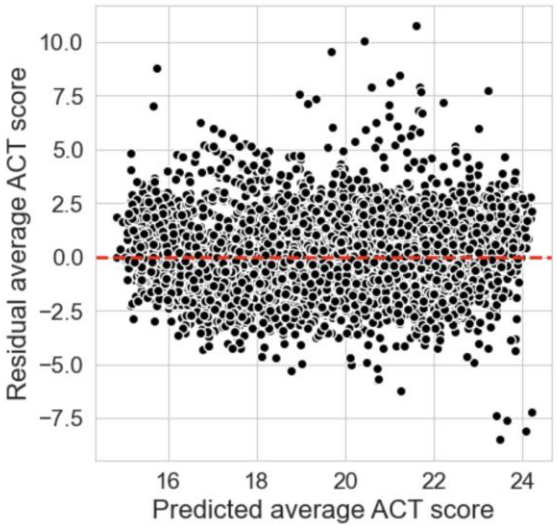


Figure 2. Residuals versus predicted average ACT score for the reduced model.

After standardizing, the free lunch coefficient was −1.78 ACT points per standard deviation, compared to 0.28 for percent college and −0.12 for unemployment. The effect of free lunch is about 6 times larger than percent college and about 14 times larger than unemployment.

Charter schools

Only 170 of the 7,227 high schools are charter schools. Their raw average ACT score is lower than non-charter schools (19.02 vs. 20.36), but they also serve more students on free or

reduced-price lunch (54% vs. 41%). After controlling for the three socioeconomic predictors, the charter coefficient was −0.18 points and not significant. Adding the charter variable did significantly improve the model overall, but this came from a third category, “Not applicable” (199 schools, 198 of them in Kentucky), which scored 0.69 points lower than expected ($p < 0.001$). The improvement in fit was also very small, with R-squared going from 0.628 to 0.630.

Discussion

These results suggest that comparing schools on raw ACT averages mostly compares the poverty levels of their students.

These findings agree with earlier research. The strong link between free or reduced-price lunch and ACT scores matches studies showing that SES is most strongly tied to achievement when it is measured at the school level and the small, non-significant charter coefficient matches national studies finding little average difference between charter and traditional public schools. The “Not applicable” charter group is almost entirely Kentucky schools. Kentucky also requires all public high school juniors to take the ACT, so its school averages include students who would not otherwise choose to take the test. This may explain why Kentucky schools scored lower than the model predicted, and it suggests that differences in state testing rules should be considered when comparing schools across states.

There are several limitations to this analysis. The data only covers 20 states and one school year, so the results may not apply to the whole country. The model was fit and evaluated on the same data, so the R-squared and MAE may be slightly optimistic for new schools. Finally, this is an observational study, so it shows that poverty is linked to lower ACT scores but cannot show that it causes them.

Conclusion

In conclusion, socioeconomic factors do predict a school’s average ACT score. My best model uses only three predictors: the percent of students on free or reduced-price lunch, the percent of adults in the area with a college degree, and the unemployment rate. Together they explain about 63% of the variation in ACT scores. The percent of students on free or reduced-price lunch is by far the most important factor. This makes sense, because it measures the poverty of the students actually attending the school, while the other measures describe the neighborhood around it. Charter schools do not score significantly differently once these factors are taken into account, so their lower raw average is mostly explained by the students they serve.